

NetTwin 2.0 Full-Stack Architecture & Multi-Stage Telemetry Pipeline

AWS VPC Substrate (5 EC2 Instances: gw, web1, app1, db1, client)

Telemetry Sources: VPC Flow Logs (10s) • CloudWatch Agent (Metrics) • VPC Traffic Mirroring (VXLAN)

Layer 1: Real-Time Ingestion & Sync Engine [ISO 23247-2] (< 1.8 ms)

sync.py (Hysteresis Sync, State Divergence Filter) • aws_adapter.py (API Ingestion & Flow Normalization)

Layer 2: Digital Twin Core & Virtual Execution [ISO 23247-3] (< 1.2 ms)

engine.py (Event Simulator) • state.py (Graph Topology State) • Counterfactual Sandbox (Branch & Evaluate)

Layer 3: Cognitive Analytics & Prescriptive Intelligence [ISO 23247-4] (< 4.4 ms)

Subspace PCA
Detector
(< 1.4 ms)

Conformal
Calibrator
(< 0.4 ms)

Drift
Monitor
(< 0.3 ms)

CausalNet-Rank
RCA
(< 1.9 ms)

Attack Graph
Risk Engine
(< 0.4 ms)

Layer 4: SafeBandit Closed-Loop Actuation [ISO 23247-5] (< 15.0 ms)

Policy Agent (Contextual Bandit) → Safety Gates (Blast Radius & SLA Check) → AWS Actuation (Boto3 API: NACL / SG)

Continuous End-to-End Budget: Ingestion (1.8ms) + Core (1.2ms) + Analytics (4.4ms) + Response (0.8ms) = 8.2 ms/tick